

# CHINMOY BHATTACHARJEE

PROFESSOR FOR MATHEMATICAL STOCHASTICS · Universität Hamburg

## PERSONAL DETAILS

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PLACE | DATE OF BIRTH: Kolkata, India | 6 December 1990  
OFFICE ADDRESS: Department of Mathematics, Universität Hamburg  
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## WORK EXPERIENCE

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SEP. 2026–present **Junior Professor (W1)** at [Department of Mathematics](#), University of Hamburg  
JUN. 2026 **Positive interim evaluation** (Habitationsäquivalent)  
Extension of Junior Professorship until AUG. 2029  
SEP. 2023–AUG. 2026 **Junior Professor (W1)** [first phase] at [Department of Mathematics](#), University of Hamburg  
AUG. 2020–AUG. 2023 **Postdoctoral Researcher** at [Department of Mathematics](#), University of Luxembourg  
AUG. 2018–JUL. 2020 Schweiz. Nationalfonds (SNF) **Postdoctoral fellow** at [Institut für Mathematische Statistik und Versicherungslehre \(IMSV\)](#), University of Bern  
JUL. 2018–AUG. 2018 **Visiting Scientist** at [Indian Statistical Institute](#), Stat-Math Unit, Kolkata

## SCIENTIFIC EDUCATION

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AUG. 2013–MAY 2018 **PhD in Applied Mathematics**, [Department of Mathematics](#), University of Southern California  
ADVISOR: Prof. Larry Goldstein  
THESIS: Stein's method and its applications in strong embeddings and Dickman approximations  
AUG. 2011–MAY 2013 **Master of Statistics (M. Stat.)**, [Indian Statistical Institute](#), Kolkata  
SPECIALIZATION: Mathematical Statistics and Probability  
AUG. 2008–MAY 2011 **Bachelor of Statistics with Honours (B. Stat. (Hons.))**, [Indian Statistical Institute](#), Kolkata

## PREPRINTS

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1. **C. Bhattacharjee**. Dickman Approximation of Randomly Weighted Sums via Stein's Method. Submitted to *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 2026+. [arXiv: 2606.27875](#)
2. **C. Bhattacharjee**, and A. Gusakova. Gaussian approximation for Extreme Points in Laguerre tessellations. Submitted to *Electronic Journal of Probability*, 2025+. [arXiv: 2510.21665](#)
3. **C. Bhattacharjee**, Y. Dhandapani, and G. Peccati. Spectra of Poisson functionals and applications to continuum percolation. Submitted to *Annals of Applied Probability*, 2024+. [arXiv: 2407.13502](#)
4. **C. Bhattacharjee**, K. Balasubramanian, W. Polonik, and Z. Shi. Multivariate Gaussian approximation in random forest via region-based stabilization. Submitted to *Annals of Applied Probability*, 2024+. [arXiv: 2403.09960](#)
5. **C. Bhattacharjee**, and R. O'Clarey. Chaos and Superconcentration for Poisson Functionals with Applications in Stochastic Geometry. Major revision in *Electronic Journal of Probability*, 2026+. [arXiv: 2603.23053](#)

## PEER-REVIEWED PUBLICATIONS

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6. **C. Bhattacharjee**, R. Versendaal, and A. Wade. Perimeter length of the convex hull of Brownian motion in the hyperbolic plane. *Applied Probability Journals*, (to appear), 2026.
7. **C. Bhattacharjee**, K. Balasubramanian, W. Polonik, and Z. Shi. Gaussian and Bootstrap Approximation for Matching-based Average Treatment Effect Estimators. *Annals of Statistics*, 54(4), 2026.
8. **C. Bhattacharjee**, and M. Schulte. Dickman approximation of weighted random sums in the Kolmogorov distance. *Annals of Applied Probability*, 35(5), 2025.
9. **C. Bhattacharjee**, I. Molchanov, and R. Turin. Central limit theorem for birth-growth model with Poisson arrivals and random growth speed. *Advances in Applied Probability*, 56(3), 2024.
10. **C. Bhattacharjee**, and I. Molchanov. Gaussian approximation for sums of region-stabilizing scores. *Electronic Journal of Probability*, 27, 2022.
11. **C. Bhattacharjee**. Gaussian approximation for rooted edges in a random minimal directed spanning tree. *Random Structures & Algorithms*, 61(3), 2022.
12. **C. Bhattacharjee**, and M. Schulte. Large degrees in scale-free inhomogeneous random graphs. *Annals of Applied Probability*, 32(1), 2022.
13. **C. Bhattacharjee**, and I. Molchanov. Convergence to scale-invariant Poisson processes and applications in Dickman approximation. *Electronic Journal of Probability*, 25, 2020.
14. **C. Bhattacharjee**, and L. Goldstein. Dickman approximation in simulation, summations and perpetuities. *Bernoulli*, 25(4A), 2019.

15. **C. Bhattacharjee**, and L. Goldstein. On Strong Embeddings by Stein's Method. *Electronic Journal of Probability*, 21, 2016.

## GRANTS

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|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–2027<br>(36 MONTHS) | <b>Superconcentration in random geometric structures</b><br>DFG Priority Program 'Random Geometric Systems' (SPP 2265)<br>ROLE: Primary Investigator<br>AMOUNT: 219,966 Euros                                                                             |
| 2026–2028<br>(24 MONTHS) | <b>Sharp noise sensitivity &amp; exceptional times in continuum percolation</b><br>Programmes for Project-Related Personal Exchange (PPP) [DST & DAAD]<br>ROLE: co-PI, together with Prof. D. Yogeshwaran (ISI, Bangalore, India)<br>AMOUNT: 11,308 Euros |

## SELECTED TALKS (IN LAST FIVE YEARS)

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| DEC. 2026<br>(upcoming) | <b>Invited talk</b> at <i>Workshop on Steins Method and Its Statistical Applications</i> , ES-SEC Business School, Cergy, France                            |
| AUG. 2026               | <b>Invited talk</b> in the Session <i>Stein-based inference</i> at European Meeting of Statisticians (EMS), Università della Svizzera italiana, Switzerland |
| JUL. 2026               | <b>Invited talk</b> at <i>Second Workshop on Steins method and networks</i> , University of Oxford, UK                                                      |
| NOV. 2025               | <b>Invited talk</b> at <i>Colloquium in Institut für Mathematik</i> , Universität Osnabrück, Germany                                                        |
| OCT. 2025               | <b>Invited talk</b> at <i>Workshop on Random geometry, graphs and extremes</i> , IUC Dubrovnik, Croatia                                                     |
| APR. 2025               | <b>Invited talk</b> at <i>Stochastics Seminar</i> , Institut for Matematik, Aarhus Universitet, Denmark                                                     |
| OCT. 2024               | <b>Invited talk</b> at <i>Workshop on random geometry and particle systems</i> , Lille (IMT Nord Europe), France                                            |
| AUG. 2024               | <b>Invited talk</b> in <i>Session on Stein's Method and Random Graphs</i> , Bernoulli-IMS World Congress, Ruhr University Bochum, Germany                   |
| JUN. 2024               | <b>Invited talk</b> at <i>Workshop on Steins method and networks</i> , University of Oxford, UK                                                             |
| JUN. 2024               | <b>Invited talk</b> at <i>Workshop on Random Geometric Systems</i> , Technische Universität Dortmund, Germany                                               |
| JUN. 2024               | <b>Contributed talk</b> at <i>Workshop on Stochastic Geometry, Stereology and Image Analysis</i> , Bad Herrenalb, Germany                                   |
| APR. 2024               | <b>Invited talk</b> at <i>Stochastics Seminar</i> , Institut für Mathematische Stochastik, University of Münster, Germany                                   |
| SEP. 2022               | <b>Invited talk</b> at <i>Mini-Workshop: Limit theorems for spatial random structures</i> , Ruhr University Bochum, Germany                                 |

JUN. 2022    **Invited talk** at *Steins Method: The Golden Anniversary*, NUS, Singapore

## TEACHING EXPERIENCE

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| UNIVERSITY OF<br>HAMBURG                                | <b>Stochastische Prozesse</b> (upcoming, in German)<br>Winter 2026/27<br><br><b>Stochastik 1 für Studierende der Informatik</b> (in German)<br>Summer 2025, Summer 2026 (jointly with Prof. Dr. Johannes Lederer)<br><br><b>Seminar on Probabilistic Methods in Combinatorics</b><br>Summer 2025, Summer 2026<br><br><b>Stochastik 2 für Studierende der Informatik</b> (in German)<br>Winter 2025/26 (jointly with Prof. Dr. Johannes Lederer)<br><br><b>Seminar on Percolation theory</b><br>Winter 2025/26<br><br><b>Introduction to Random Graphs</b><br>Master in Mathematics; Winter 2023/24, Winter 2024/25<br><br><b>Introduction to Stein's Method</b><br>Master in Mathematics; Summer 2024 |
| UNIVERSITY OF<br>LUXEMBOURG                             | <b>Probability and Statistics II</b><br>BMATH fourth semester; Spring 2023, Spring 2022<br><br><b>Refresher in Probability</b><br>Master of Data Science first semester; Fall 2022, Fall 2021<br><br><b>Probabilistic Methods in Combinatorics</b><br>BMATH fifth semester; Fall 2021, Fall 2020                                                                                                                                                                                                                                                                                                                                                                                                      |
| UNIVERSITY OF<br>BERN                                   | <b>Introduction to Random Graphs</b><br>MMATH second semester; Spring 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| UNIVERSITY OF<br>SOUTHERN<br>CALIFORNIA<br>(ASSISTANCE) | <b>Math 125 - Calculus I</b> ; Fall 2017<br><b>Math 308 - Statistical Inference and Data Analysis II</b> ; Spring 2016<br><b>Math 307 - Statistical Inference and Data Analysis I</b> ; Fall 2015<br><b>Math 226 - Calculus III</b> ; Fall 2014, Spring 2015                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## SUPERVISION

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| PHD.                 | Rowan O'Clarey (from Oct. 2024 – Sep. 2027 Expected)<br>-Funded by DFG grant no. 531540467<br>TOPIC: Superconcentration in random geometric structures |
| OTHER<br>SUPERVISION | Amélie Boussemart (Master's thesis)<br>Oliver Gertke (Master's thesis)<br>Maika Lu Beecken (Bachelor's thesis)                                         |

## ORGANIZATION & ACADEMIC SERVICE

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### *Workshop Organization:*

- MAR. 2025    **Session Co-chair:** *Spatial stochastics, disordered media, and complex networks*, in the *German Probability and Statistics Days*, Dresden, Germany.  
Co-chaired with B. Jahnel (TU Braunschweig/WIAS Berlin)
- SEP. 2024    **Workshop Coorganizer:** *Workshop on Stein's method and stochastic geometry*, Hamburg, Germany.  
Coorganized with M. Schulte (TUHH, Hamburg)

### *Peer Reviewing:*

- JOURNALS    Annals of Statistics (AoS), Applied Probability Journals, Bernoulli, Extremes, Electronic Journal of Probability (EJP), Journal of the American Statistical Association (JASA), Probability Theory & Related Fields (PTRF), Random Structures & Algorithms (RSA), Statistics & Probability Letters, Stochastic Processes & their Applications (SPA)

## LANGUAGE SKILLS

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Bengali (mother tongue), English (C2), German (B1), French (A1), Hindi (C1)